

Planning and Reinforcement Learning in AI Systems

(PARLAI)

Advanced Topics in RL

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Summary

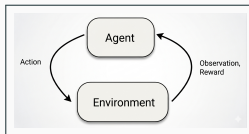
Single Agent Autonomy

Summary

Approaches to
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What we haven't
covered

What does it mean to be autonomous?



We seek a **policy** from states (or knowledge about the state) to actions.

Approaches to Decision Making

We seek a **policy**, which we mark as π that maps states $\mathcal{S}$ or (knowledge about the state) to actions $\mathcal{A}$.

Deterministic $\pi : \mathcal{S} \mapsto \mathcal{A}$ or probabilistic $\pi : \mathcal{S} \times \mathcal{A} \mapsto [0, 1]$

More accurately, we seek a mapping from beliefs $\mathcal{B}$, representing knowledge about the current state (formally defined later on), to actions.

Deterministic $\pi : \mathcal{B} \mapsto \mathcal{A}$ or probabilistic $\pi : \mathcal{B} \times \mathcal{A} \mapsto [0, 1]$

Basic approaches:

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Approaches to Decision Making

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Basic approaches:

- Programming/reactive
- Model-Based Planning
- Machine Learning (from data and experience)
- Reinforcement Learning

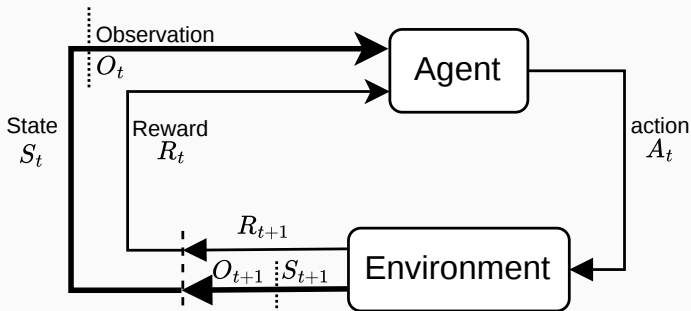
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Modeling the Dynamics of the Environment

- Deterministic vs. stochastic
- Full vs. partial observability



Summary

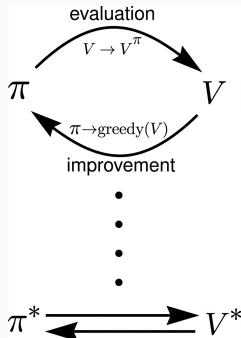
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Policy Evaluation and Policy Optimization

Prediction / Evaluation: evaluate the future given a policy -
evaluate the quality of a given policy

Control/ Update / Improvement: optimize the policy.



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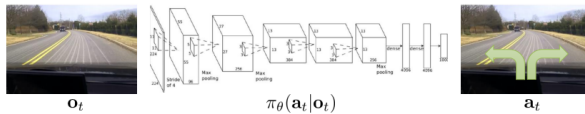
What we haven't
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DM as Learning

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DM as Learning



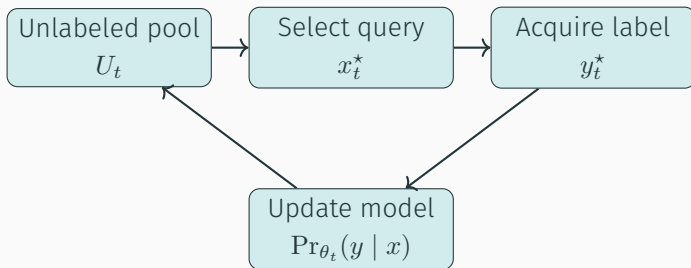
Is this the same observation?

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Active Learning Loop



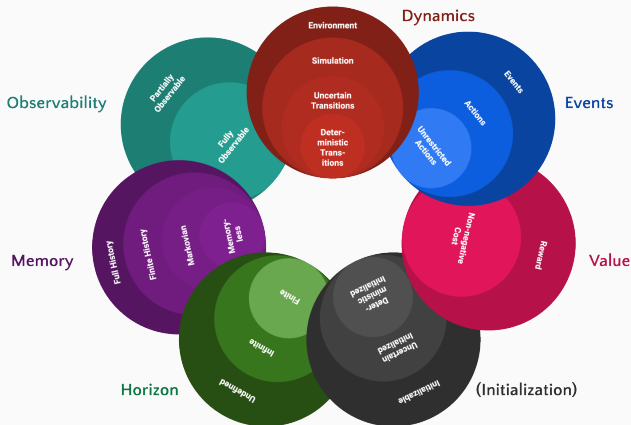
The hypothesis space over the underlying model is represented by θ .

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Many Forms of Planning



<https://github.com/fteicht/icaps24-skdecide-tutorial/tree/main>

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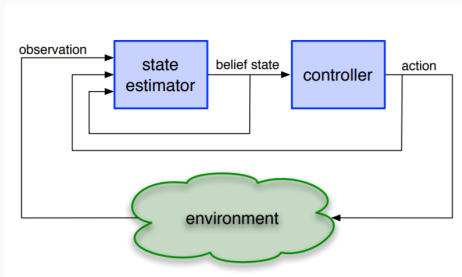
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Planning in Belief Space

Two key challenges when planning in belief space:

- **Belief tracking** - what is the state of the world ?
- **Policy computation** - what is the best action to perform ?



Pineau, Nicholas and Thrun. "A hierarchical approach to POMDP planning and execution." 2001. <https://www.cs.mcgill.ca/~jpineau/files/jpineau-icml01.pdf>

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MCTS is not just for games

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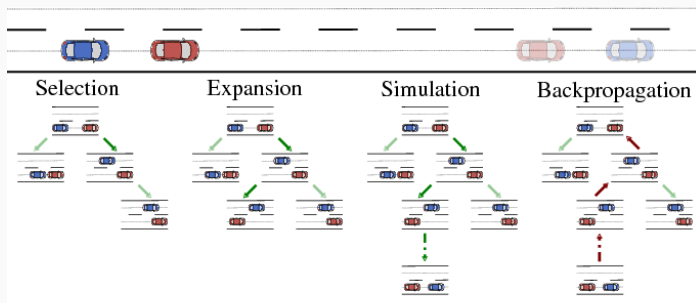
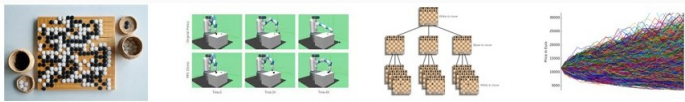


Fig. 1: Phases of Monte Carlo Tree Search for an overtaking maneuver; the

<https://www.semanticscholar.org/paper/Decentralized-Cooperative-Planning-for-Automated-Kurzer-Zhou/585b73322365ba2d0afa7449691c81cb98777599>

Simulation (Rollout)

- A simulation (also called playout or rollout) chooses moves according to some policy, until reaching a terminal state
- Various options for simulation policy (e.g., random)
- The value of a state is estimated as the average utility over a number of simulations



Used in many applications and in different algorithms.

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Simulators



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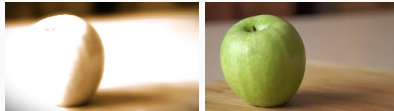
Partial and noisy sensing

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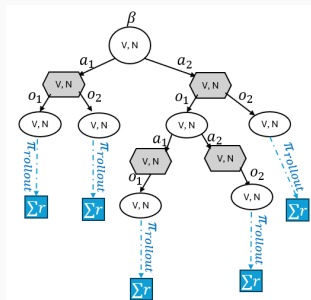
Partial and noisy sensing



Partial observability over time



- Extending MCTS to POMDPs.
- Relevant to large or continuous POMDPs where explicit belief updates are intractable.
- Basic idea: use Monte-Carlo in two ways:
 - sampling start states from the belief.
 - sampling histories using a black box simulator.
- **Online algorithm:** iteratively expands a search tree of histories.



Example

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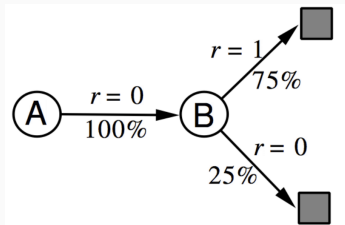


What to do next?

AB Example

Two states A, B; no discounting; 8 episodes of experience

- A, 0, B, 0
- B, 1
- B, 1
- B, 1
- B, 1
- B, 1
- B, 1
- B, 0



What is $v(A)$, $v(B)$?

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How do we switch from optimality equations to update formulas?

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$$\mathcal{V}^*(s) = \max_a \mathbb{E}_{\pi^*} [G_t | S_t = s, A_t = a]$$

How can the optimality equation be used within a solution algorithm ?

How do we switch from optimality equations to update formulas?

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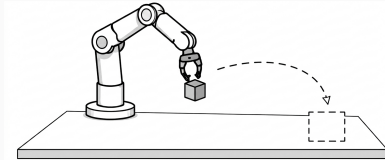
$$\mathcal{V}^*(s) = \max_a \mathbb{E}_{\pi^*} [G_t | S_t = s, A_t = a]$$

How can the optimality equation be used within a solution algorithm ?

$$\mathcal{V}_{k+1}(s) = \max_a \mathbb{E} [G_t | S_t = s, A_t = a]$$

$$\mathcal{V}(s) \leftarrow \mathcal{V}(S) + \alpha(G - \mathcal{V}(S))$$

Example

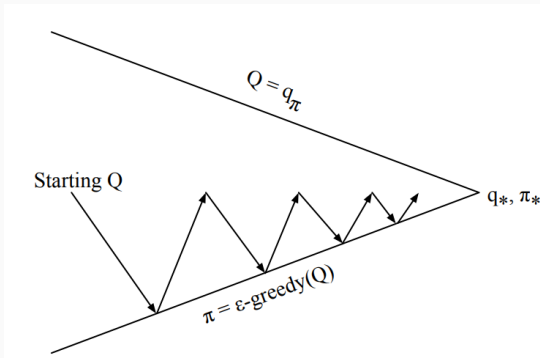


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Monte-Carlo Control



Every Iteration:

- Policy evaluation: Monte-Carlo policy evaluation. $Q \approx q_\pi$
- Policy improvement: ϵ -greedy improvement

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Temporal-Difference (TD) Control

Do local updates of utility/value function on a per-action basis.

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Temporal-Difference (TD) Control

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Do local updates of utility/value function on a per-action basis.

What does TD learn from?

Temporal-Difference (TD) Control

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Do local updates of utility/value function on a per-action basis.

What does TD learn from? TD methods learn directly from episodes of experience

Model-free or model-based?

Temporal-Difference (TD) Control

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Do local updates of utility/value function on a per-action basis.

What does TD learn from? TD methods learn directly from episodes of experience

Model-free or model-based? TD is model-free: no learning of MDP transitions / rewards (doesn't try to estimate entire transition function).

Is there a need to wait for the episode to end?

Temporal-Difference (TD) Control

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Do local updates of utility/value function on a per-action basis.

What does TD learn from? TD methods learn directly from episodes of experience

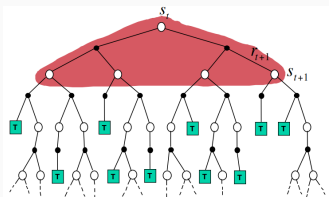
Model-free or model-based? TD is model-free: no learning of MDP transitions / rewards (doesn't try to estimate entire transition function).

Is there a need to wait for the episode to end? TD learns from incomplete episodes, by **bootstrapping** - updates a guess towards a guess.

Monte-Carlo vs. Temporal-Difference vs. Dynamic Programming Backup

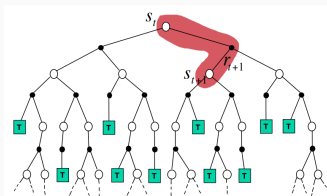
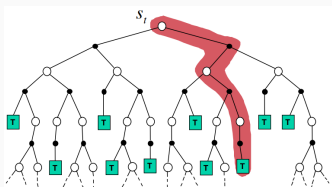
Which is which ?

$$V(s_t) \leftarrow \mathbb{E}_{\pi}[R_{t+1} + \gamma V(s_{t+1})]$$



$$V(s_t) \leftarrow V(s_t) + \alpha(G_t V(s_t))$$

$$V(s_t) \leftarrow V(s_t) + \alpha(R_{t+1} + \gamma V(s_{t+1}) - V(s_t))$$



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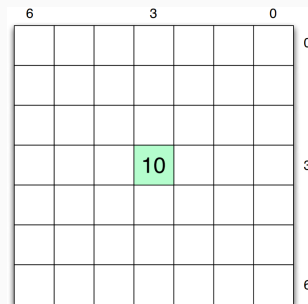
REINFORCE

- 1: Initialize policy parameters θ
 - 2: **while** not converged **do**
 - 3: Generate an episode using π_θ
 - 4: **for** each time step t in the episode **do**
 - 5: Compute return $G_t = \sum_{k=0}^{T-t-1} \gamma^k r_{t+k+1}$
 - 6: $\theta \leftarrow \theta + \alpha G_t \nabla_\theta \log \pi_\theta(a_t | s_t)$
 - 7: **end for**
 - 8: **end while**
-

But What If..

Grid with no obstacles, deterministic actions Up-Down-Left-Right, no discounting, -1 reward everywhere except +10 at goal.

- Features for $s = (x, y)$:
 $f_1(s) = x$, $f_2(s) = y$
 $f_3(s) = |3 - x| + |3 - y|$
- $V_\theta(s) = \theta_0 + \theta_1 x + \theta_2 y + \theta_3 f_3(s)$
- Is there a good linear approximation?
- Yes!
- $\theta_0 = 10, \theta_1 = \theta_2 = 0, \theta_3 = -1$



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Model-based RL Motivation

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Model-based RL exploits a learned model to predict the consequences of actions, enabling planning and generalization to new situations with fewer real-world interactions.



Although in practice the line between these two techniques can become blurred, as a coarse guide it is useful for dividing up the space of algorithmic possibilities.

Approaches to model-based control?

Sarah Keren

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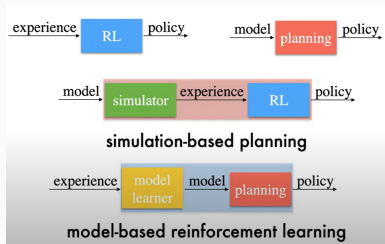
What we haven't
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Approaches to model-based control?

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By Michael Littman <https://youtu.be/45FKxa3qPHo?si=chpp9SRxzjvKRZb9>

Use the model to:

- **Create data:** generate imagined experience for learning.
- **Plan:** search, dynamic programming, or MPC.
- **Improve a policy or value function:** train the agent using model-based predictions.

Model Learning: Design Decisions

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covered

Model Representation: What model class should we use?

table, linear model, Gaussian process, neural network, etc.

Model Output: Should the model predict a value or a distribution?

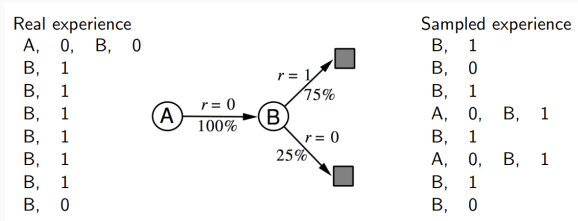
Measuring Loss: What loss function should we use?

MSE, negative log-likelihood, KL divergence, etc.

There are many ways to answer these questions; we will cover only some possible solutions

Sample-Based DM: Back to the AB Example

- Construct a table-lookup model from real experience
- Apply model-free RL to sampled experience



e.g., Monte-Carlo learning: $V(A) = 1, V(B) = 0.75$

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Use the Model to Expedite Learning

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Another way to use a learned model is for generating additional training data.

Real experience is expensive:

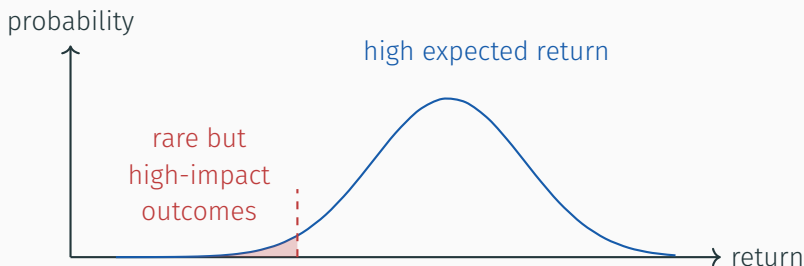
$$(s, a, r, s') \sim \text{environment}.$$

Simulated experience (i.e., data generated from the learned model) is cheaper:

$$(\tilde{s}, \tilde{a}, \tilde{r}, \tilde{s}') \sim \hat{M}.$$

The Gap: Expected Return Is Not Enough

Most standard RL (and multi-agent RL) algorithms optimize expected cumulative reward.



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Approaches to
Reliability

What we haven't
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In high-stakes systems, a policy can have good average performance while still producing unacceptable failures.

Approaches to Reliability

Safe RL, Robust RL, and Related Settings

Different extensions ask for reliability in different ways.

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What we haven't
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Safe RL, Robust RL, and Related Settings

Different extensions ask for reliability in different ways.

Safe RL focuses on constraint violations.

Can the agent learn and act without violating safety constraints?

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Safe RL, Robust RL, and Related Settings

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Can the agent learn and act without violating safety constraints?

Robust RL and Bayesian RL focus on uncertainty in the environment.

Does the policy still work under model errors, disturbances, or distribution shift?

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Does the policy still work under model errors, disturbances, or distribution shift?

Risk-sensitive RL focuses on the tail of the outcome distribution.
How bad is performance in the worst cases?

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Dimensions of Multi-Agent Systems

- **Control:** from centralized to decentralized.
- **Decision-making approach:** planning, learning, RL
- **Communication:** from no communication to limitless communication
- **Observability:** no sensing (conformant) to full information
- **Shared resources/interaction:** space, energy, etc.
- **Objectives/utilities:** from collaborative to adversarial

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What we haven't covered

What We Haven't Covered

Throughout this course we focused on the foundations of sequential decision making:

- Classical planning
- Planning under uncertainty
- Reinforcement learning
- Safe and robust decision making
- Multi-agent AI (just a little)

Modern AI contains many additional directions that build upon these ideas.

Today these areas increasingly overlap and are often combined within the same intelligent system.

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Many Important Topics Remain

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Among the topics we did not have time to explore are:

- Hierarchical RL
- Meta-learning
- Continual learning
- Transfer learning
- Curriculum learning

Many current research papers combine several of these ideas simultaneously.

Beyond a Single Intelligent Agent

Many real systems contain multiple autonomous decision makers.

Examples include

- autonomous vehicles,
- energy markets,
- distributed sensor networks.

These settings require reasoning about

- cooperation,
- competition,
- communication,
- negotiation,
- strategic behavior.

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Approaches to
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Modern AI increasingly moves beyond digital environments.

Physical AI integrates

- perception,
- planning,
- reinforcement learning,
- reasoning,
- control,
- interaction with humans,
- execution in the physical world.

This requires handling

- uncertainty,
- real-time constraints,
- safety,
- imperfect sensing,
- changing environments.

The Journey Continues

Decision making under uncertainty remains one of the central challenges of AI.

Future intelligent systems will increasingly combine

Planning Learning Reasoning Perception
Interaction

to operate reliably in complex real-world environments.

The goal is no longer simply to build agents that solve problems, but intelligent systems that continually learn, adapt, collaborate, and assist humans in the real world.

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